Agile Approach

Rationale for Agile Adoption

- **Evolving requirements** due to **complex** Sanskrit grammar rules
- Need for **iterative validation** with linguists
- **Continuous feedback** from research stakeholders

Framework Used

- Scrum **(2-week sprints)** for **development & testing**
- Integrated with **XP** practices for quality

Sprint Structure (Project Context)

- **Sprint Duration: 2 Weeks**
- **Activities:**
  - Requirement refinement (with linguists)
  - Feature development (NLP modules)
  - Testing & validation
  - Review with stakeholders

Scrum Framework

Workflow in This Project

- **Product Backlog: NLP features (morphology, parsing, API)**
- **Sprint Planning:** Prioritization based on **MoSCoW**
- **Sprint Backlog:** Selected tasks for each sprint
- **Sprint (2 Weeks):** Development + testing cycle

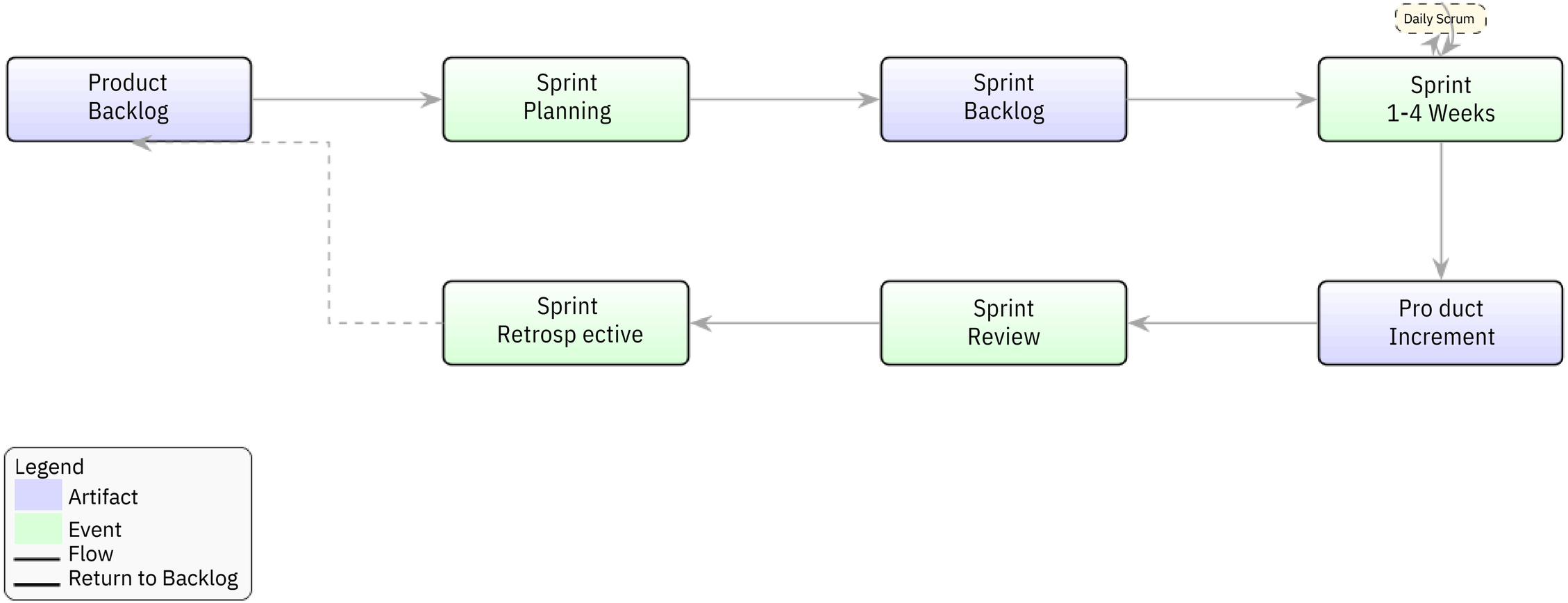
Execution Cycle

- **Daily Scrum → progress tracking**
- **Sprint Review → stakeholder validation (linguists)**
- **Sprint Retrospective → process improvement**
- **Output → Incremental feature delivery**

Project-Specific Insight

- Ensures continuous validation of linguistic accuracy
- Supports iterative delivery of NLP modules
- Improves stakeholder collaboration

Scrum Framework



Sprint Progress Tracking

Sprint Context

- **Sprint Duration: 10 Days (2 Weeks)**
- **Initial Workload: 50 Story Points**
- **Scope: NLP module development + testing**

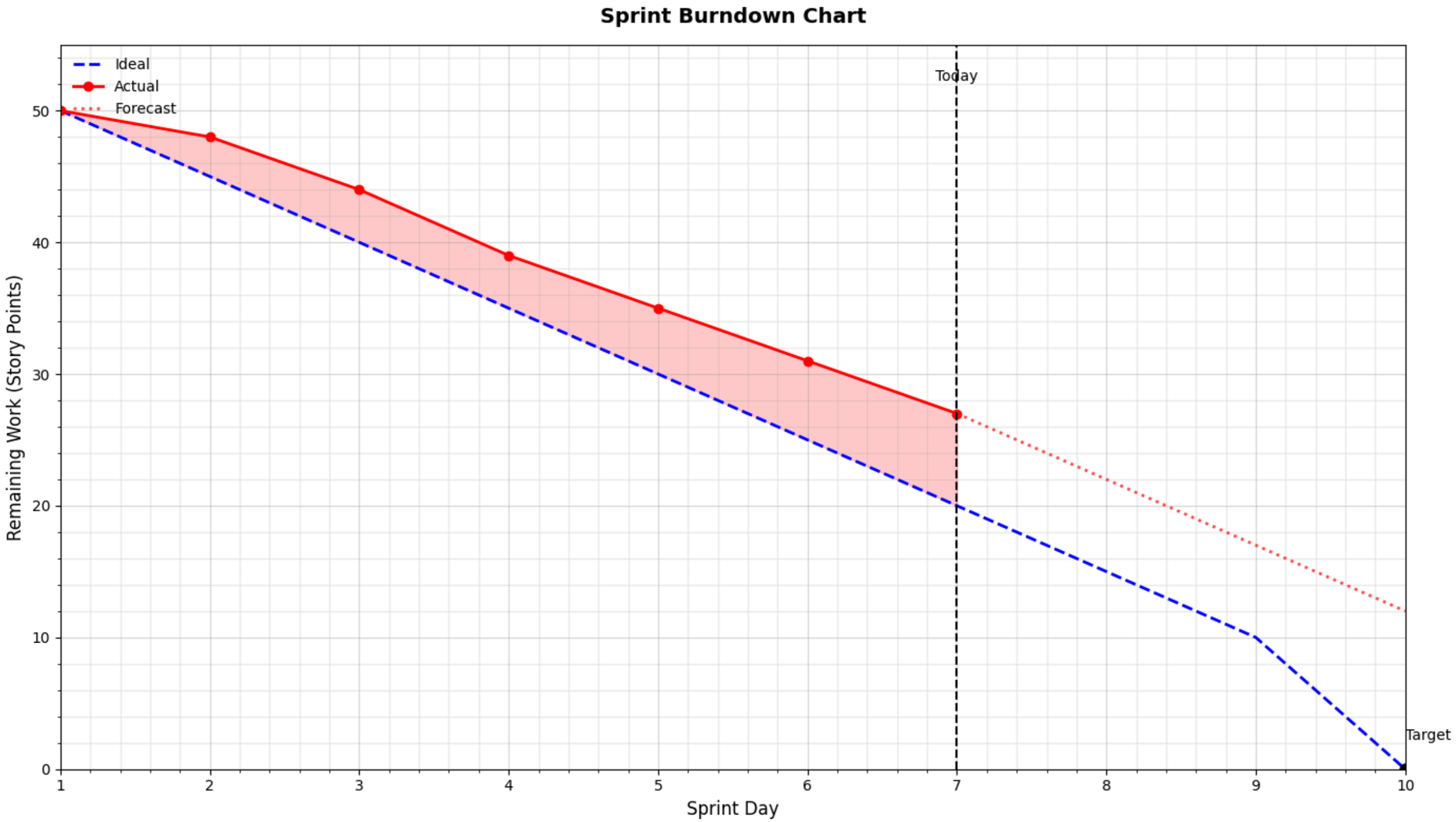
Performance Observation

- Actual progress remains above ideal line
- Indicates slower completion rate

Forecast Insight

- Remaining work at sprint end → ~12 story points
- Risk of spillover to next sprint

Management Action

- Reallocate resources for critical tasks
- Improve task breakdown in next sprint
- Increase monitoring through daily scrums

Sprint Progress Tracking